

Quantum Mechanics-II Foundations Notes

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1. Observables in Quantum Mechanics

Definition

An observable is any measurable physical quantity and is represented by a **Hermitian operator**.

Observable $\longleftrightarrow$ Hermitian Operator

Examples:

- Position: $\hat{x} = x$
- Momentum: $\hat{p} = -i\hbar \frac{d}{dx}$
- Energy: $\hat{H}$
- Angular momentum: $\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}$

Why Hermitian?

- Eigenvalues are real (measurements must be real).
- Eigenstates form orthonormal basis.
- Ensures probabilistic interpretation.

2. Eigenvalues and Measurement

If

$$\hat{A}|a\rangle = a|a\rangle$$

Then:

- a is a possible measurement outcome.
- After measurement, the system collapses to $|a\rangle$.

Expectation Value

$$\langle A \rangle = \langle \psi | \hat{A} | \psi \rangle$$

Represents the average result over many measurements.

3. Completeness Condition

Definition

A set of states $\{|n\rangle\}$ is complete if any state can be expanded as:

$$|\psi\rangle = \sum_n c_n |n\rangle$$

Completeness Relation

$$\sum_n |n\rangle \langle n| = I$$

Continuous Case

$$\int |x\rangle \langle x| dx = I$$

Meaning

A complete basis can describe all possible quantum states.

4. Orthonormality vs Completeness

Orthonormality

$$\langle m | n \rangle = \delta_{mn}$$

Completeness

$$\sum_n |n\rangle \langle n| = I$$

Both are required for a valid basis.

5. Commutators and Compatibility

Commutator

$$[A, B] = AB - BA$$

Meaning

- $[A, B] = 0 \Rightarrow$ observables compatible.
- $[A, B] \neq 0 \Rightarrow$ observables incompatible.

Physical Interpretation

Order of measurement matters if operators do not commute.

6. Simultaneous Eigenkets

If

$$[A, B] = 0$$

Then there exist states $|\psi\rangle$ such that:

$$A|\psi\rangle = a|\psi\rangle, \quad B|\psi\rangle = b|\psi\rangle$$

Meaning

Both observables can have definite values simultaneously.

Example

$$[L^2, L_z] = 0 \Rightarrow |l, m\rangle \text{ is a simultaneous eigenket.}$$

7. Uncertainty Principle

General Form

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|$$

Cases

Commuting observables:

$$[A, B] = 0 \Rightarrow \Delta A \Delta B \geq 0$$

Simultaneous precision is possible.

Non-commuting observables:

$$[A, B] \neq 0 \Rightarrow \Delta A \Delta B > 0$$

Simultaneous precision forbidden.

Example

$$[x, p] = i\hbar \Rightarrow \Delta x \Delta p \geq \frac{\hbar}{2}$$

8. Root Cause of Uncertainty

- Definite value requires eigenstate.
- Non-commuting operators do not share eigenstates.
- Therefore, simultaneous definite values are impossible.

Key Insight

Uncertainty is a property of the **state**, not measurement error.

9. Measurement Disturbance vs Fundamental Uncertainty

Incorrect: Measurement disturbance causes uncertainty.

Correct: No quantum state exists with definite values of both observables.

Measurement disturbance is a consequence of collapse.

10. Anticommutators

Definition

$$\{A, B\} = AB + BA$$

Role

- Appears in fermions and Dirac theory.
- Does NOT determine uncertainty.

Uncertainty depends only on the commutator.

11. Why $i\hbar$ Appears

- $\hbar$ is the quantum of action.
- i ensures Hermitian operators and real observables.
- Momentum operator structure requires it.

12. Deep Structural Insight

- Commutator measures incompatibility.
- Non-commuting operators correspond to different axes in Hilbert space.
- Quantum numbers exist because commuting observables exist.

13. Diagnostic Questions

1. Why must wavefunctions be complex?
2. What does an eigenvalue represent physically?
3. Why can position and momentum not be known simultaneously?
4. What does $|\langle\phi|\psi\rangle|^2$ represent?
5. Why does completeness imply the identity operator?
6. Why is orthonormality alone insufficient?
7. Why does perturbation theory require completeness?
8. Why can energy and angular momentum be known together in hydrogen?
9. Why can L_x and L_y not be definite simultaneously?
10. Why do quantum numbers arise from commuting operators?

14. One-Sentence Core Truths

- Observables are Hermitian operators.
- Eigenvalues are measurement outcomes.
- Completeness ensures all states can be described.
- Commutation determines compatibility.
- Uncertainty arises from lack of common eigenstates.
- Measurement disturbance is not the root cause.